Ho-min Park

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ABOUT ME

Postdoctoral researcher in Computer Science Engineering specializing in **biomedical and biological AI**, with extensive experience developing deep learning solutions for medical imaging (laparoscopy, MRI, microscopy), structural biology (CRISPR-Cas systems and anti-CRISPR proteins), and genomics. Demonstrated expertise in building interpretable, clinically oriented ML pipelines that have been validated in collaboration with surgeons, biologists, and environmental scientists. Author of peer-reviewed publications in high-impact venues including the *Journal of Clinical Oncology*, *International Journal of Surgery*, *IEEE Transactions on Affective Computing*, and *Transactions on Machine Learning Research*, holder of 2 patents, and contributor to multiple open-source bioinformatics tools. Currently a postdoctoral researcher at Ghent University Global Campus and incoming Assistant Professor at the Texas Children's Hospital Data Center (Houston, TX), where research will focus on advancing biomedical AI for pediatric and oncological applications.

Citation metrics (Google Scholar, May 2026): 295 total citations; h-index 10; i10-index 10.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher November 2025 - Present
Ghent University Global Campus (GUGC), Incheon, South Korea

- ✓ Conducting research on biomedical and biological AI under the supervision of Prof. Joris Vankerschaver.
- ✓ Contributing to multi-institutional collaborations in surgical AI (peritoneal metastasis detection, laparoscopic video analysis) and genomics (data augmentation for genomic deep learning).

Assistant Professor (Incoming) 2026
Texas Children's Hospital Data Center, Houston, TX, USA

- ✓ Joining as Assistant Professor to lead research on machine learning for pediatric and oncological applications.

SKILLS

Computational
Python & Deep Learning Frameworks

- ✓ Primary programming language, with expertise in AI model development using PyTorch and PyTorch Lightning.
- ✓ Extensive experience implementing deep learning models for medical imaging, genomics, and multimodal data, focusing on MLOps tasks and integrating with Weights and Biases (WandB).

GitHub

- ✓ Extensively used for version control and project sharing, with experience in collaborative development and documentation.
- ✓ Actively maintained multiple public repositories for AI research projects and educational materials.

Weights and Biases (WandB)
- ✓ Actively used for experiment tracking and model performance monitoring in MLOps tasks.

Languages

Korean
- ✓ Native speaker.

English
- ✓ *Writing:* Demonstrated through technical documentation and research publications.
- ✓ *Reading:* Evident through the comprehension of technical and academic literature.
- ✓ *Speaking and Listening:* Applied in collaborative environments with international teams.

FUNDED PROJECTS

Depression Diagnosis & Drug Adherence Research (2022 - 2025)
- ✓ Developed advanced models for the MuSe-Stress and MuSe-Personalization challenges.
- ✓ Implemented Transformer Encoders and novel feature engineering techniques.
- ✓ Achieved 2nd place in MuSe-Personalisation 2023 and 3rd place in MuSe-Stress 2022.
- ✓ Contributed to the field of affective computing with interpretable AI models.
- ✓ **Funding:** National Research Foundation of Korea (Grant No.: NRF-2020K1A3A1A68093469), Department of Biotechnology in India (Grant No.: DBT/IC-12031(22)-ICD-DBT), and Ghent University Global Campus

Environmental Monitoring Project (2020 - 2023)
- ✓ Led development of MP-Net, an AI system for automated microplastics detection.
- ✓ Designed and implemented a microplastics annotation tool for dataset creation.
- ✓ Managed collaboration between environmental scientists and AI engineers.
- ✓ **Funding:** Special Research Fund of Ghent University (Grant No.: 01N01718) and Ghent University Global Campus

Smart Packaging System Development (2018 - 2021)
- ✓ Developed and patented two AI-powered measurement systems for industrial conveyor belts.
- ✓ Led implementation of 3D deep learning-based item classification system.
- ✓ Designed integrated size and type monitoring system.
- ✓ Collaborated with industry partners to deploy solutions in real factory environments.
- ✓ **Funding:** Ministry of Trade, Industry, and Energy (MOTIE), Korea (Project No.: 10077285)

PATENTS

- ✓ Park, H., & De Neve, W. (2020) An apparatus and method for measuring the size of high-speed objects and classifying their type on conveyors using light curtain sensors. Korean Intellectual Property Office. Application no.: 10-2020-0045098, Registration no.: 10-2273758.
- ✓ Park, H., & De Neve, W. (2018) Apparatus and method for measuring the size of high-speed boxes on conveyors using an RGB-D camera. Korean Intellectual Property Office. Application no.: 10-2018-0148118, Registration no.: 10-2066862.

PUBLICATIONS

Citation counts in italics are from Google Scholar (May 2026). First-author publications are marked with first-author position. Full publication list and metrics available at Google Scholar.

Journal Articles

- ✓ Tozzi, F., **Park, H. M.**, Mousavi, S. A., Van Liefferinge, M., Moon, D., Fadaei, S., et al. (2026) Multimodal machine learning for staging laparoscopy: a combined image analysis and morphologic tool for the discrimination of peritoneal metastasis. *International Journal of Surgery*, **112**(1): 373-383.
- ✓ Tozzi, F., Mousavi, S. A., Van Liefferinge, M., Moon, D., **Park, H.**, Fadaei, S., et al. (2024) Development of a video-based deep learning model for differentiation of malignant and benign lesions during staging laparoscopy: Is the machine better than the expert? *Journal of Clinical Oncology*, **42**(16_suppl): e13616. *(1 citation)*
- ✓ Tozzi, F., **Park, H. M.**, Moon, D., Mousavi, S. A., Van Liefferinge, M., Fadaei, S., et al. (2024) Machine learning models for classification and morphological assessment of peritoneal lesions during staging laparoscopy. *European Journal of Surgical Oncology*, **50**.
- ✓ Lee, H., Ozbulak, U., **Park, H.**, Depuydt, S., De Neve, W., & Vankerschaver, J. (2024) Assessing the reliability of point mutation as data augmentation for deep learning with genomic data. *BMC Bioinformatics*, **25**(1): 170. *(11 citations)*
- ✓ **Park, H.** (first author), Kim, G., Oh, J., Van Messem, A., & De Neve, W. (2024) Interpreting stress detection models using SHAP and attention for MuSe-Stress 2022. *IEEE Transactions on Affective Computing*, **15**(01): 1-17. DOI: 10.1109/TAFFC.2024.3488112. *(9 citations)*
- ✓ Ozbulak, U., Lee, H. J., Boga, B., Anzaku, E. T., **Park, H.**, Van Messem, A., et al. (2023) Know your self-supervised learning: A survey on image-based generative and discriminative training. *Transactions on Machine Learning Research (TMLR)*. *(71 citations)*
- ✓ **Park, H.** (first author), Won, J., Park, Y., Anzaku, E. T., Vankerschaver, J., Van Messem, A., De Neve, W., & Shim, H. (2023) CRISPR-Cas-Docker: Web-based in silico docking and machine learning-based classification of crRNAs with Cas proteins. *BMC Bioinformatics*, **24**(1): 167. Springer. *(5 citations)*
- ✓ **Park, H.** (first author), Park, Y., Berani, U., Bang, E., Vankerschaver, J., Van Messem, A., De Neve, W., & Shim, H. (2022) In silico optimization of RNA-protein interactions for CRISPR-Cas13-based antimicrobials. *Biology Direct*, **17**(1). DOI: 10.1186/s13062-022-00339-5. *(13 citations)*
- ✓ **Park, H. M.** (first author), Park, Y., Vankerschaver, J., Van Messem, A., De Neve, W., & Shim, H. (2022) Rethinking protein drug design with highly accurate structure prediction of anti-CRISPR proteins. *Pharmaceuticals*, **15**(3): 310. DOI: 10.3390/ph15030310. *(25 citations)*
- ✓ **Park, H.** (first author), Park, S., de Guzman, M. K., Baek, J. Y., Cirkovic Velickovic, T., Van Messem, A., & De Neve, W. (2022) MP-Net: Deep learning-based segmentation for fluorescence microscopy images of microplastics isolated from clams. *PLoS ONE*, **17**(6): e0269449. DOI: 10.1371/journal.pone.0269449. *(32 citations)*

Conference Articles

- ✓ Mousavi, S. A., Tozzi, F., **Park, H.**, Anzaku, E. T., Van Liefferinge, M., Rashidian, N., et al. (2024) A Reference-Based Approach for Tumor Size Estimation in Monocular Laparoscopic Videos. *International Workshop on Computational Mathematics Modeling in Cancer*. Springer. *(2 citations)*
- ✓ Singh, A. K., Kim, G., Kim, J., **Park, H.**, Choi, B. J., & De Neve, W. (2025) RAMM: A Residual Attention Multimodal Model for Humor Detection. *International Conference on Intelligent Human Computer Interaction*, 229-240. *(3 citations)*
- ✓ Nguyen, K. T., **Park, H.**, Oh, G., Vankerschaver, J., & De Neve, W. (2025) Towards Improved Cervical Cancer Screening: Vision Transformer-Based Classification and Interpretability. *IEEE 22nd International Symposium on Biomedical Imaging (ISBI)*, 1-5. *(5 citations)*
- ✓ **Park, H.** (first author), Yun, I., Kim, M., Nguyen, K. T., Van Messem, A., & De Neve, W.

(2024) Interpretable Rotator Cuff Tear Diagnosis Using MRI Slides with CAMscore and SHAP. *Medical Imaging 2024: Computer-Aided Diagnosis*, Vol. 12927: 683-693. SPIE.
- ✓ Singh, A. K., Kim, G., Kim, J., **Park, H.**, Choi, B. J., & De Neve, W. (2024) Exploring Multimodal Approaches and Fusion Methods for CEO Social Attribute Prediction in 2024 MuSe-Perception. *Proceedings of the 5th on Multimodal Sentiment Analysis Challenge and Workshop: Social Perception and Humor*, 36–44. Melbourne VIC, Australia: ACM. DOI: 10.1145/3689062.3689081. *(2 citations)*
- ✓ **Park, H. M.** (first author), Kim, G., Van Messem, A., & De Neve, W. (2023) MuSe-Personalization 2023: Feature engineering, hyperparameter optimization, and transformer-encoder re-discovery. *Proceedings of the 4th Multimodal Sentiment Analysis Challenge and Workshop: Mimicked Emotions, Humour and Personalisation*, 89–97. Ottawa ON, Canada: ACM. DOI: 10.1145/3606039.3613104. *(5 citations)*
- ✓ **Park, H.** (first author), Yun, I., Kumar, A., Singh, A. K., Choi, B. J., Singh, D., & De Neve, W. (2022) Towards multimodal prediction of time-continuous emotion using pose feature engineering and a transformer encoder. *Proceedings of the 3rd International on Multimodal Sentiment Analysis Workshop and Challenge*, 47–54. DOI: 10.1145/3551876.3554807. *(13 citations)*
- ✓ Kang, H., **Park, H.**, Ahn, Y., Van Messem, A., & De Neve, W. (2021) Towards a quantitative analysis of class activation mapping for deep learning-based computer-aided diagnosis. *Medical Imaging 2021: Image Perception, Observer Performance, and Technology Assessment*, Vol. 11599, SPIE. DOI: 10.1117/12.2580819. *(18 citations)*
- ✓ **Park, H.** (first author), Kang, B., Van Messem, A., & De Neve, W. (2021) 3-D deep learning-based item classification for belt conveyors targeting packaging and logistics. *Pattern Recognition, ICPR International Workshops and Challenges, Proceedings, Part VI*, 12666: 578–591. Cham: Springer. DOI: 10.1007/978-3-030-68799-1_42. *(2 citations)*
- ✓ Baek, J. Y., de Guzman, M. K., **Park, H.**, Park, S., Shin, B., Cirkovic Velickovic, T., Van Messem, A., & De Neve, W. (2021) Developing a segmentation model for microscopic images of microplastics isolated from clams. *Pattern Recognition, ICPR International Workshops and Challenges, Proceedings, Part VI*, 12666: 86–97. Cham: Springer. DOI: 10.1007/978-3-030-68780-9_9. *(4 citations)*
- ✓ Kim, M., **Park, H.**, Kim, J. Y., Kim, S. H., Hoeke, S., & De Neve, W. (2020) MRI-based diagnosis of rotator cuff tears using deep learning and weighted linear combinations. *Machine Learning for Healthcare Conference (MLHC)*, 292-308. *(19 citations)*
- ✓ **Park, H.** (first author), Van Messem, A., & De Neve, W. (2020) Item measurement for logistics-oriented belt conveyor systems using a scenario-driven approach and automata-based control design. *IEEE 7th International Conference on Industrial Engineering and Applications (ICIEA 2020)*, Bangkok and Phuket. *(8 citations)*
- ✓ **Park, H. M.** (first author), Van Messem, A., & De Neve, W. (2019) Box-Scan: An efficient and effective algorithm for box dimension measurement in conveyor systems using a single RGB-D camera. *The 7th IIAE International Conference on Industrial Application Engineering*, Kitakyushu, Japan. *(16 citations)* **[Best Presentation Award]**

AWARDS

- ✓ **Second Place**, MuSe-Personalisation Challenge 2023 — international competition at ACM Multimedia, for contributions to multimodal sentiment analysis.
- ✓ **Third Place**, MuSe-Stress Challenge 2022 — international competition at ACM Multimedia, for contributions to multimodal sentiment analysis.
- ✓ **Best Presentation Award**, 7th IIAE International Conference on Industrial Application Engineering, 2019.

EDUCATION

Ph.D. in Computer Science Engineering 2018 - 2025

Ghent University, Belgium

- ✓ Dissertation: "Advancing Detection through Deep Learning for Industrial, Environmental, and Health Applications"
- ✓ Advisors: Prof. Wesley De Neve, Prof. Arnout Van Messem

M.S. in Computer Engineering 2016 - 2018

Ajou University, Suwon, South Korea

- ✓ Thesis: "Cited count prediction modeling from large citation network"

B.S. in Computer Science and Engineering 2010 - 2016

Ajou University, Suwon, South Korea

- ✓ Major in Computer Science and Engineering

MENTORING AND EDUCATIONAL LEADERSHIP

Founder and Lead Instructor, AI Vacation School (AIVS)

- ✓ Founded and led the *AI Vacation School (AIVS)* at the Center for Biosystems and Biotech Data Science, Ghent University Global Campus, as a volunteer initiative to provide intensive AI education to undergraduate students from 2021 to 2026, supporting over 30 participants across summer and winter cohorts. [Program Repository, Lecture Slides]
- ✓ Mentored participants who have advanced to top graduate programs and research positions, including KAIST, Seoul National University, UNIST, Ghent University (Belgium), and Scripps Research Institute (San Diego, CA, USA).
- ✓ Co-authored peer-reviewed publications with alumni in venues such as *IEEE Transactions on Affective Computing*, *BMC Bioinformatics*, *PLoS ONE*, *SPIE Medical Imaging*, and *ACM Multimodal Sentiment Analysis Workshop*, demonstrating the program's effectiveness in producing research-active students.
- ✓ Developed and released open-source educational and research artifacts in collaboration with participants, including *CRISPR-Cas-Docker* (web service), *MP-Net* segmentation toolkit, *Microplastics Annotation Package*, and the *MPSet* dataset on Kaggle.
- ✓ Presented the program at the *IGC Research Showcase 2025, AI in Education* session, sharing pedagogical insights with the four universities of Incheon Global Campus (SUNY Korea, University of Utah Asia Campus, George Mason University Korea, Ghent University Global Campus). [Presentation]
- ✓ Provided extensive individualized mentoring, including portfolio development guidance, reference letter writing for graduate school and industry applications, and ongoing career advising for alumni.

REFERENCES

References available upon request.